

Vision-based Approach for robust UAV Detection:



- **Introduction:**

- Unmanned Aerial Vehicles (UAVs), commonly known as drones, are widely used in civilian and military applications such as surveillance, delivery systems, mapping, and disaster monitoring.
- However, the misuse of drones can pose serious security threats, including unauthorized surveillance, intrusion into restricted airspaces (airports, defense bases, government facilities), and transportation of illegal or harmful payloads.
- Therefore, reliable drone detection systems are essential. In this project, we aim to study and evaluate modern deep learning–based detection models for identifying UAVs at distances of up to **2 km**.

- **Motivation:**

- Security agencies require reliable drone detection systems that can **quickly and accurately identify UAVs** to enable timely action against potential threats.
- Detecting drones is challenging because they often appear **very small in images and operate in complex environments such as open skies or urban areas or in forest area**.
- Therefore, there is a need to study and evaluate **advanced deep learning–based detection models** that can reliably detect UAVs at long distances (up to **2 km**) under real-world conditions.

Existing work / Literature review:



- This project continues the work previously carried out by a senior student on UAV detection using YOLOv9, and their results are used as a reference for further analysis and comparison with other modern detection models.
- In previous work, UAV detection was performed using the **YOLOv9** deep learning model, which was used to detect drones from aerial images.
- After detecting the drone region using YOLOv9, the detected image was further classified using an **EfficientNet-B3 classifier** to improve accuracy and reduce false detections.

- **Gaps in the existing solutions/work:**

- The existing approach performs well when the drone is **close and clearly visible**, but its accuracy decreases when the drone is **very far away or appears very small**, especially in **cluttered environments such as clouds, trees, or urban backgrounds**.

- A. **Distance-based accuracy**

- **Close range** (large boxes): $\approx 0.93 - 1.00$
 - **Medium range** (moderate boxes): $\approx 0.90 - 0.96$
 - **Very far / tiny targets** (very small boxes): $\approx 0.70 - 0.78$

- B. **Environment-based accuracy**

- **Bright / clear sky**: $\approx 0.98 - 1.00$
 - **Night / low light** : $\approx 0.85 - 0.98$
 - **Cluttered / ambiguous backgrounds (forests, urban)**: $\approx 0.70 - 0.80$

Work Done till date:



- **Details of proposed work/solutions:**

- I find out different modern object detection models and understand their working principles and suitability for long-range drone detection.

Model	Key Features	limitation
YOLOv9	CNN detector, uses GELAN and PGI for featural earning	Good detection but still depends on NMS and struggles with very small distant UAVs
YOLO11	Anchor-free detector, dual-head prediction, uses DFL	Good balance of speed and accuracy, suitable for small object detection
YOLO26	NMS-free detection, STAL for small targets	New model but still under study and requires experimentation
YOLO-NAS	Architecture optimized using NAS for high accuracy	Anchor-based detection may not perform well for tiny UAV targets
RT-DETR	Transformer-based detector	Computationally heavier and may not be ideal for UAV detection
NSSRD	Uses noise suppression and super-resolution before detection	Mainly preprocessing pipeline, not a full standalone detection model
ED-NET	Lightweight model designed for small object detection	May sacrifice accuracy compared to larger models

- **Result achieved (if any):**

- Currently, I am in the learning and analysis phase of different UAV detection models. Based on the understanding so far, YOLO11 or a customized version of YOLO11 appears to be a promising approach, and further work will focus on improving accuracy by incorporating UAV-specific features.